

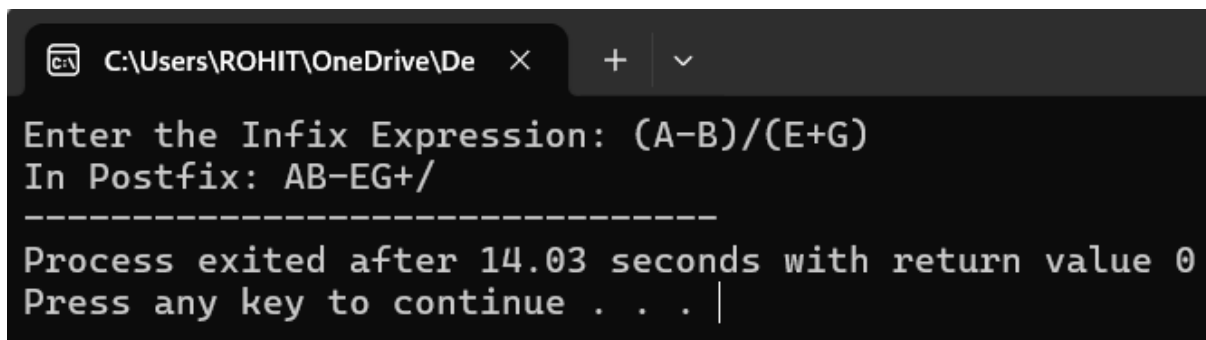
Aim: Implement program for i.Infix to Postfix conversion, ii.Infix to Prefix conversion, iii.Postfix Evaluation, iv.Prefix Evaluation.

1.Infix to Postfix conversion.

CODE:

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;
int operatorPrecedence(char op){
    if(op == '^'){
        return 3;
    }
    if(op == '*' || op == '/'){
        return 2;
    }
    if(op == '+' || op == '-'){
        return 1;
    }
    return -1;
}
string convertInfixToPostfix(string s){
    stack<char> st;
    string res;
    int sz = s.size();
    for(int i = 0; i < sz; i++){
        char c = s[i];
        if(isalnum(c)){
            res += c;
        }
        else if (c == '('){
            st.push(c);
        }
        else if(c == ')'){
            while(!st.empty() && st.top() != '('){
                res = res+st.top();
                st.pop();
            }
            if(!st.empty()){
                st.pop();
            }
        }
        else{
            while(!st.empty() && operatorPrecedence(s[i]) <= operatorPrecedence(st.top())){
                res = res+st.top();
                st.pop();
            }
            st.push(c);
        }
    }
}
```

```
while(!st.empty()){
    res = res+st.top();
    st.pop();
}
return res;
}
int main(){
    string exp;
    cout<<"Enter the Infix Expression: ";
    getline(cin,exp);
    cout<<"In Postfix: "<<convertInfixToPostfix(exp);
    return 0;
}
```

OUTPUT:

```
C:\Users\ROHIT\OneDrive\De  X  +  v

Enter the Infix Expression: (A-B)/(E+G)
In Postfix: AB-EG+/
-----
Process exited after 14.03 seconds with return value 0
Press any key to continue . . . |
```

2.Infix to Prefix conversion.**CODE:**

```
#include <iostream>
#include <bits/stdc++.h>
using namespace std;
int operatorPrecedence(char op){
    if(op == '^'){
        return 3;
    }
    if(op == '*' || op == '/'){
        return 2;
    }
    if(op == '+' || op == '-'){
        return 1;
    }
    return -1;
}
string convertInfixToPostfix(string s){
    stack<char> st;
    string res;
    int sz = s.size();
    for(int i = 0;i<sz;i++){
        char c = s[i];
        if(isalnum(c)){
            res += c;
```

```

    }
    else if (c == '('){
        st.push(c);
    }
    else if(c == '){
        while(!st.empty() && st.top() != '('){
            res = res+st.top();
            st.pop();
        }
        if(!st.empty()){
            st.pop();
        }
    }
    else{
        while(!st.empty() && operatorPrecedence(s[i]) <= operatorPrecedence(st.top())){
            res = res+st.top();
            st.pop();
        }
        st.push(c);
    }
}
while(!st.empty()){
    res = res+st.top();
    st.pop();
}
return res;
}
string convertToPrefix(string infix){
    reverse(infix.begin(), infix.end());
    for(int i = 0;i<infix.size();i++){
        if(infix[i]=='('){
            infix[i]=')';
        }
        else{
            if(infix[i]==')'){
                infix[i]='(';
            }
        }
    }
    string postfix = convertInfixToPostfix(infix);
    reverse(postfix.begin(), postfix.end());
    return postfix;
}
int main(){
    string exp;
    cout<<"Enter the Infix Expression: ";
    getline(cin,exp);
    cout<<"In Prefix: "<<convertToPrefix(exp);

```

```

        return 0;
    }

```

OUTPUT:

```

C:\Users\ROHIT\OneDrive\De  X  +  v

Enter the Infix Expression: (A+B)*(G-H)
In Prefix: *+AB-GH
-----
Process exited after 19.02 seconds with return value 0
Press any key to continue . . . |

```

3. Postfix Evaluation.**CODE:**

```

#include <iostream>
#include <stack>
#include <bits/stdc++.h>
using namespace std;
int scanNum(char ch){
    int value;
    value = ch;
    return int(value - '0');
}
int isOperator(char ch){
    if(ch=='+' || ch=='-' || ch=='*' || ch=='/' || ch=='^')
        return 1;
    return -1;
}
int isOperand(char ch){
    if(ch>='0' && ch<='9')
        return 1;
    return -1;
}
float operation(int a, int b, char op){
    if(op=='+')
        return a+b;
    if(op=='-')
        return b-a;
    if(op=='*')
        return a*b;
    if(op=='/')
        return b/a;
    if(op=='^')
        return pow(b,a);
    else
        return INT_MIN;
}
float postfixExp(string postfix){

```

```

    int a,b;
    stack<float> st;
    string::iterator it;
    for(it = postfix.begin(); it != postfix.end(); it++){
        if(isOperator(*it) != -1){
            a=st.top();
            st.pop();
            b=st.top();
            st.pop();
            st.push(operation(a,b,*it));
        }
        else{
            if(isOperand(*it)>0){
                st.push(scanNum(*it));
            }
        }
    }
    return st.top();
}
int main(){
    string exp;
    cout<<"Enter Postfix expression: ";
    getline(cin,exp);
    cout<<"Result: "<<postfixExp(exp);
    return 0;
}

```

OUTPUT:

```

C:\Users\ROHIT\OneDrive\De
Enter Postfix expression: 9 8 * 3 +
Result: 75
-----
Process exited after 13.41 seconds with return value 0
Press any key to continue . . . |

```

4. Prefix Evaluation.**CODE:**

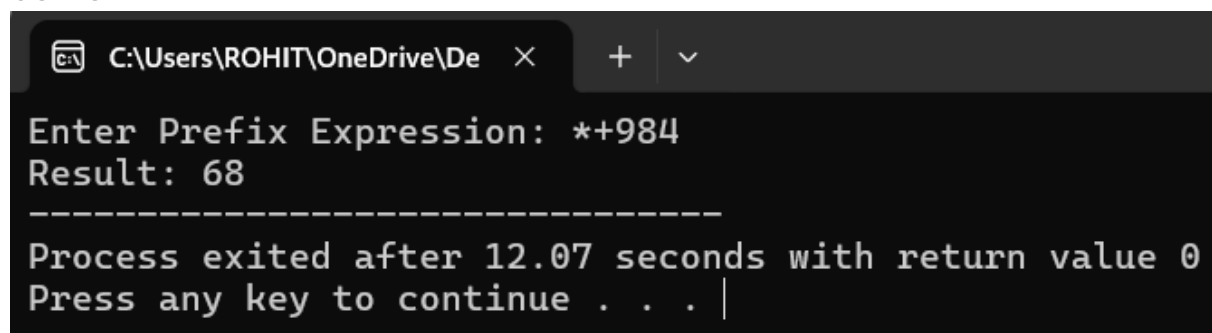
```

#include <iostream>
#include <stack>
#include <bits/stdc++.h>
using namespace std;
bool isOperand(char ch){
    return isdigit(ch);
}
double evaluatePrefix(string exp){
    stack<double> st;
    for(int j=exp.size()-1;j>=0;j--){

```

```
        if(isOperand(exp[j])){
            st.push(exp[j] - '0');
        }
        else{
            double a = st.top();
            st.pop();
            double b = st.top();
            st.pop();
            switch(exp[j]){
                case '+':
                    st.push(a+b);
                    break;
                case '-':
                    st.push(a-b);
                    break;
                case '*':
                    st.push(a*b);
                    break;
                case '/':
                    st.push(a/b);
                    break;
                case '^':
                    st.push(pow(a,b));
                default:
                    cout<<"Invalid Operation!";
            }
        }
    }
    return st.top();
}

int main(){
    string exp;
    cout<<"Enter Prefix Expression: ";
    getline(cin,exp);
    cout<<"Result: "<<evaluatePrefix(exp);
    return 0;
}
```

OUTPUT:

```
C:\Users\ROHIT\OneDrive\De  ×  +  ∨

Enter Prefix Expression: *+984
Result: 68
-----
Process exited after 12.07 seconds with return value 0
Press any key to continue . . . |
```